Yanjun Chen 陈彦筠

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Research Interests

Reinforcement Learning for Large Language Models and Embodied Agents: models are trainable; the environments that train them are not. Toward making the environment trainable, the way models are.

Education

The Hong Kong Polytechnic University, Hong Kong — Sept 2024 to present

PhD, Department of Computing. Candidature confirmed June 2026.
Advisors: Prof. Wenjie Li (Maggie), Prof. Wei Zhang. Joint doctoral training with the Eastern Institute of Technology (EIT), Ningbo.

Jinan University, Guangzhou, China — Sept 2020 to Jun 2024

B.Eng. in Cyberspace Security. GPA 88.6 / 100 · Rank 6 of 62 · Outstanding Graduate.

Experience

Microsoft Research Asia, Singapore — Sept 2026 to present

Research Intern.

Publications

First-Author

1. **Yanjun Chen**, Jinghan Wang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **Action Shaping: Policies Absorb What They Can Express.** *arXiv:2609.32752*, 2026 (in submission).
2. **Yanjun Chen**, Yirong Sun, Hanlin Wang, Jinghan Wang, Xinming Zhang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **The Trace Is the State: Exact Credit Assignment for LLM Agent Teams.** *arXiv:2603.06859*, 2026 (in submission).
3. **Yanjun Chen**, Yirong Sun, Xinghao Chen, Jian Wang, Xiaoyu Shen, Wenjie Li, Wei Zhang. **Integrating Chain-of-Thought for Multimodal Alignment: A Study on 3D Vision-Language Learning.** *arXiv:2503.06232*, 2025. [dataset]
4. **Yanjun Chen**, Xinming Zhang, Xianghui Wang, Zhiqiang Xu, Xiaoyu Shen, Wei Zhang. **Rethinking Soft Actor-Critic in High-Dimensional Action Spaces: The Cost of Ignoring Distribution Shift.** *arXiv:2410.16739*, 2024.
5. **Yanjun Chen**, Dawei Zhu, Yirong Sun, Xinghao Chen, Wei Zhang, Xiaoyu Shen. **The Accuracy Paradox in RLHF: When Better Reward Models Don't Yield Better Language Models.** *EMNLP 2024*.

Co-Authored

1. Jun Xue, Junze Wang, Shanze Wang, Xinming Zhang, **Yanjun Chen**, Wei Zhang. **FastDSAC: Unlocking the Potential of Maximum Entropy RL in High-Dimensional Humanoid Control.** *NeurIPS 2026*.
2. Xinghao Chen, Anhao Zhao, Heming Xia, Xuan Lu, Hanlin Wang, **Yanjun Chen**, Wei Zhang, Jian Wang, et al. **Reasoning Beyond Language: A Comprehensive Survey on Latent Chain-of-Thought Reasoning.** *Findings of EMNLP 2026*.
3. Yuhan Hu, Yirong Sun, **Yanjun Chen**, Xinghao Chen, Xiaoyu Shen, Wei Zhang. **Breaking the Pre-Planning Barrier: Adaptive Real-Time Coordination of Heterogeneous UAVs.** *IEEE Transactions on Vehicular Technology*, 2026.
4. Jinghan Wang, Gaoliang Peng, **Yanjun Chen**, Wei Zhang, Wentao Wu, Tianchen Liu. **Auditable progressive multimodal bearing fault diagnosis with Large language model Screening.** *Measurement*, 2026.
5. Yunuo Liu, Dawei Zhu, Zena Al-Khalili, Dai Cheng, **Yanjun Chen**, Dietrich Klakow, Wei Zhang, Xiaoyu Shen. **PricingLogic: Evaluating LLMs Reasoning on Complex Tourism Pricing Tasks.** *EMNLP 2025*.
6. Xinghao Chen, Zhijing Sun, Wenjin Guo, Miaoran Zhang, **Yanjun Chen**, Yirong Sun, Hui Su, Yijie Pan, et al. **Unveiling the Key Factors for Distilling Chain-of-Thought Reasoning.** *Findings of ACL 2025*.

7. Yirong Sun, Dawei Zhu, **Yanjun Chen**, Erjia Xiao, Xinghao Chen, Xiaoyu Shen. **Fine-Grained and Multi-Dimensional Metrics for Document-Level Machine Translation.** *NAACL 2025 SRW*.

Additional co-authored preprints, 2024 to 2026; full list on Google Scholar.

Open Source

C3 github.com/EIT-EAST-Lab/C3

Official implementation of *The Trace Is the State* (arXiv:2603.06859): exact per-decision credit by counterfactual continuation, and evaluation of any credit signal by its bias, variance, and agreement with an independent reference.

AccuracyParadox-RLHF github.com/EIT-NLP/AccuracyParadox-RLHF

Official implementation of *The Accuracy Paradox in RLHF* (EMNLP 2024): reproducible training pipelines, evaluation harnesses, and reference reward models.

3D-CoT dataset huggingface.co/datasets/Battam/3D-CoT

Chain-of-thought annotations for 3D vision-language alignment (arXiv:2503.06232).

Myco, independent project github.com/Battam1111/Myco

Agent infrastructure in Rust: persistent memory and knowledge management for AI agents in the Model Context Protocol (MCP) ecosystem.

Technical Skills

Post-Training	SFT and RLHF; PPO, DPO, GRPO; reward modeling and evaluation; RLVR; SAC and off-policy RL for continuous control.
Training Infra	DeepSpeed (ZeRO), Megatron-LM; verl, OpenRLHF, Hugging Face TRL; LoRA / PEFT; multi-GPU distributed training; vLLM inference.
Frameworks	PyTorch, Hugging Face Transformers / Datasets / Accelerate; Weights & Biases.
Programming	Python, C/C++, Bash; Git, Linux.
Languages	Mandarin (native), English (professional), Japanese (elementary).

Teaching & Service

Teaching (TA)	COMP5221 Software Project Management · COMP1002 Computational Thinking and Problem Solving · COMP1411 Introduction to Computer Systems · COMP5567 Distributed Algorithms and Protocols for Blockchains (2024 to 2026).
Reviewing	ACL Rolling Review (ARR) 2026 · IEEE Transactions on Neural Networks and Learning Systems (TNNLS).